

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS DEPARTMENT

YEAR 11 2 UNIT ASSESSMENT TASK 2

JULY 2002

Time Allowed: 70 minutes

Name: _____

Teacher: _____

Instructions:

- Show all necessary working
- Marks may be withheld awarded for careless or badly arranged work
- Calculators may be used
- Marks shown are a guide and may be varied
- Hand in this question paper on top of your correctly ordered solutions
- Start each question on a new page.

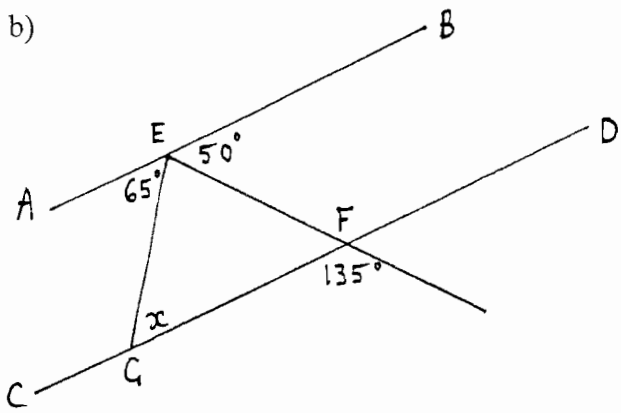
Question	1	2	3	4	5	6	7	8	Total
Marks									

Question 1

Marks

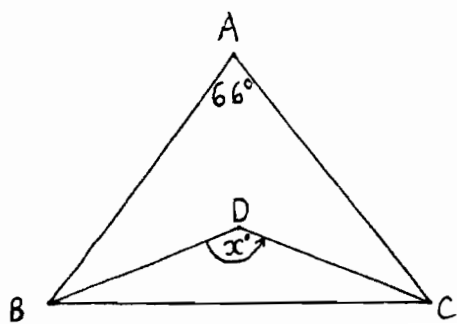
- a) (i) For the function $y = \frac{1}{x+1}$, write down the value of x at which the function doesn't exist. As x approaches $\pm \infty$, y approaches which number? 2
- (ii) Find the y intercept 1
- (iii) Hence sketch the curve showing the asymptotes and y intercept 3
- b) Solve for $0 \leq x \leq 90^\circ$: 2
 $\sin 60^\circ = \cos (x+10)^\circ$

Question 2

- a) Shade the region on the number plane satisfying:
 $x^2 + y^2 \leq 4$ and $y < 2x$ 3
- b)  2
- (i) Find x giving reasons 2
- (ii) Hence deduce whether $AB \parallel CD$ 1
- c) Evaluate $\sin(-135)^\circ$ leaving your answer in exact form 2

Question 3

- a) (i) Sketch $y = |x-2|$ and $y = 3$ on the same number plane diagram 2
- (ii) Hence or otherwise solve $|x-2| = 3$ 2

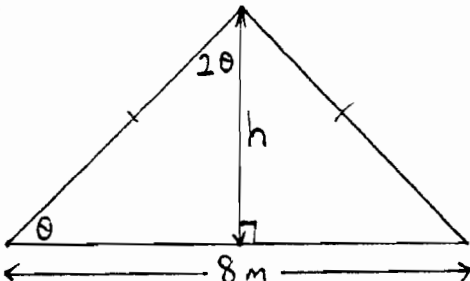
- b)  3
- In $\triangle ABC$, DB bisects $\angle ABC$ and DC bisects $\angle ACB$.
 Copy the diagram onto your answer sheet showing all the above information and find the value of x giving reasons.

Question 4

a) A regular polygon has each interior angle equal to 150° . Find the sum of all its interior angles. 2

b) Simplify $(\sec \theta - 1)(\sec \theta + 1)$ 2

c)



(i) Find the size of θ to the nearest degree 1

(ii) Find the length h correct to one decimal place 2

Question 5

a) If A is acute and $\tan A = \frac{5}{3}$, draw a right angled triangle and find

(i) the exact values of $\sin A$ and $\cos A$ 2

(ii) Hence show that $\sin A = \sqrt{1 - \cos^2 A}$ 2

b) Prove the identity $\frac{\cos \theta}{1 + \sin \theta} = \sec \theta (1 - \sin \theta)$ 3

Question 6

Let A and B be the points $(0, 1)$ and $(2, 3)$ respectively

(i) Find the coordinates of the midpoint of AB 1

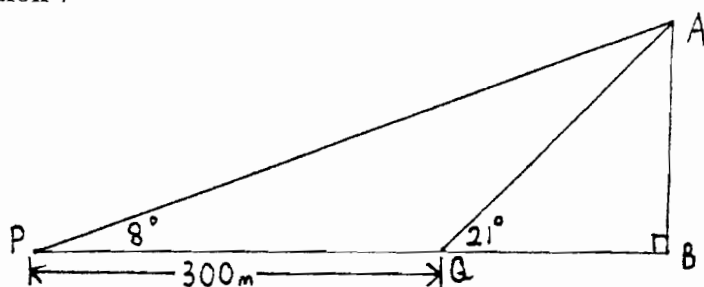
(ii) Find the slope of the line AB 1

(iii) Find the equation of the perpendicular bisector of AB 3

(iv) The point P lies on the line $y = 2x - 9$ and is equidistant from A and B . Using your answer to part (iii) or otherwise, find the coordinates of P . 2

Question 7

a)



From a position P , Claudia finds that the angle of elevation of the top A , of a billboard AB is 8° . After walking 300m directly towards the billboard to the point Q she finds that the angle of elevation of A is 21° .

(i) Copy the diagram onto your answer sheet and find $\angle PAQ$. 1

(ii) Calculate the length of AQ (nearest metre) 2

(iii) Find the height of the billboard (nearest metres) 2

b) Solve $\tan \theta = \frac{-1}{\sqrt{3}}$ if $0 \leq \theta \leq 360^\circ$ 2

Question 8

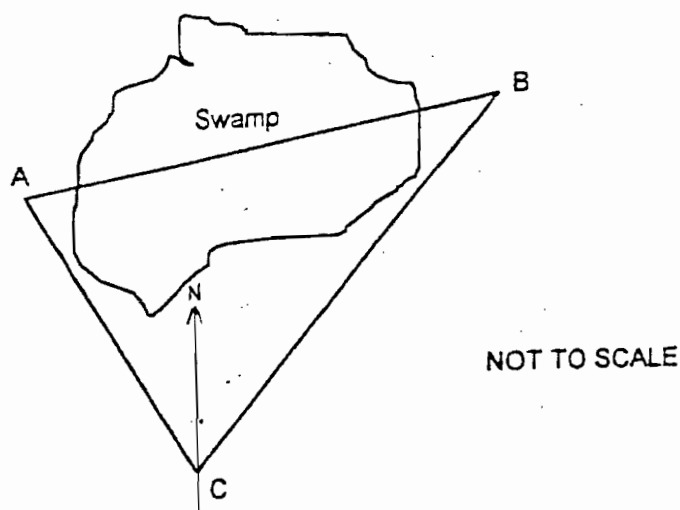
a) Find the shortest distance between the parallel lines 3

$$3x - 4y + 7 = 0 \quad \text{and}$$

$$3x - 4y - 3 = 0$$

b) Find the exact value of $\operatorname{cosec} 300^\circ$ 2

c) 3



The diagram shows two points A and B on opposite sides of a crocodile infested swamp. From a point C a surveyor notes that the bearings to A and B are 340° and 036° respectively. The distances AC and BC were then measured and found to be 180 metres and 212 metres respectively.

(i) Copy the diagram showing this information.

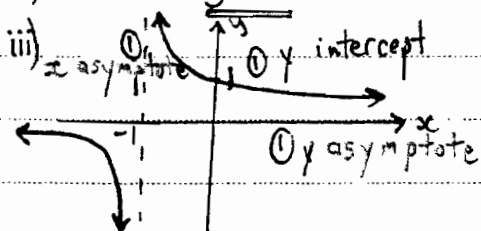
(ii) Calculate the distance from A to B across the swamp (nearest metre).

QUESTION 1

a) i) $x = -1$ ①

$y \rightarrow 0$ ①

ii) $x = 0$ $y = 1$ ①

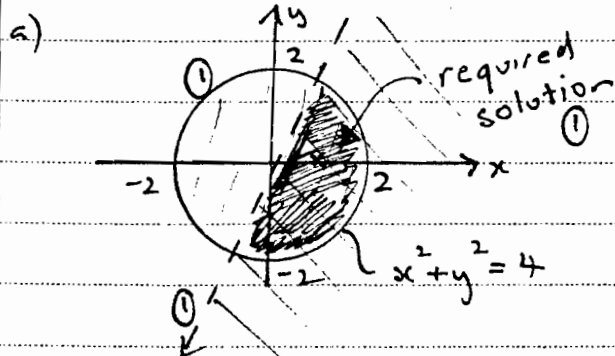


b) $\sin 60^\circ = \cos (x + 10)^\circ$

$\therefore x + 10 + 60 = 90^\circ$ ①

$x = 20^\circ$ ①

QUESTION 2



$y < 2x$ test (1,0)

b) i)

$\hat{FEG} = 65^\circ$ (supp L's)

$\angle EFG = 45^\circ$ (supp L's) ①

$x = 70^\circ$ ($\angle$ sum of $\Delta = 180^\circ$) ①

ii) AB not $\parallel$ CD

(alt. L's not equal) ①

c) $\sin (-135)^\circ = \sin 225^\circ$ ①

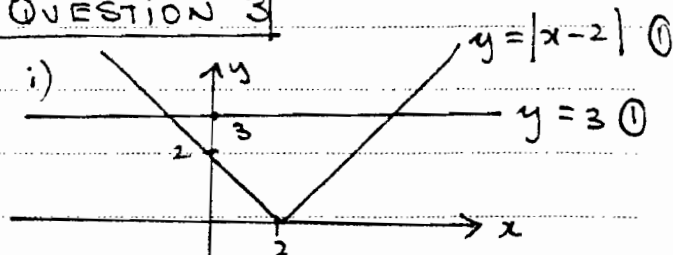
$= \sin (180 + 45)^\circ$

$= -\sin 45$

$= -1/\sqrt{2}$ ①

QUESTION 3

a) i)

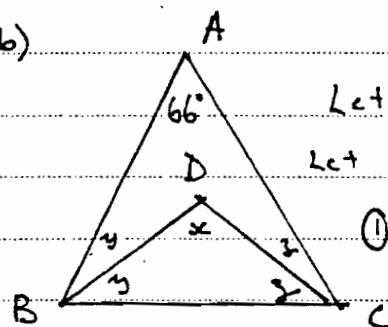


ii) $x - 2 = 3$ $x - 2 = -3$

$x = 5$ ①

$x = -1$ ① or from sketch

b)



Let $\hat{ABD} = \hat{DBC} = y$

Let $\hat{ACD} = \hat{DCB} = z$

$2y + 2z = 114^\circ$ (angle sum Δ)

$y + z = 57^\circ$ ①

$\therefore x = 123^\circ$ (angle sum ΔBDC) ①

QUESTION 4

a) int angle $= 150^\circ$

$\therefore$ ext angle $= 30^\circ$ ①

$\frac{360}{30} = 12 \therefore 12$ sided polygon

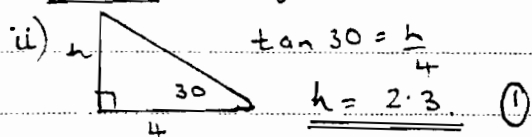
Int angle sum $= 1800^\circ$ ①

b) $(\sec \theta - 1)(\sec \theta + 1)$

$= \sec^2 \theta - 1$ ①

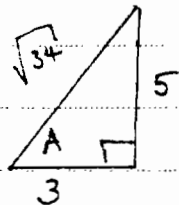
$= \tan^2 \theta$ ①

c) i) $\theta = 30^\circ$ (angle sum Δ) ①



QUESTION 5

a) $\tan A = \frac{5}{3}$



i) $\sin A = \frac{5}{\sqrt{34}}$ ①

$\cos A = \frac{3}{\sqrt{34}}$ ①

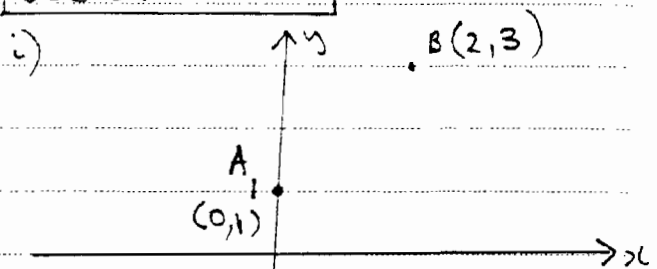
ii) $RHS = \sqrt{1 - \cos^2 A}$
 $= \sqrt{1 - \left(\frac{3}{\sqrt{34}}\right)^2}$
 $= \sqrt{1 - \frac{9}{34}}$ ①
 $= \sqrt{\frac{25}{34}}$
 $= \frac{5}{\sqrt{34}}$ ①

= LHS

b)

$LHS = \frac{\cos \theta}{1 + \sin \theta}$
 $= \frac{\cos \theta}{1 + \sin \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$ ①
 $= \frac{\cos \theta (1 - \sin \theta)}{1 - \sin^2 \theta}$
 $= \frac{\cos \theta (1 - \sin \theta)}{\cos^2 \theta}$ ①
 $= \frac{1 - \sin \theta}{\cos \theta}$
 $= \sec \theta (1 - \sin \theta)$ ①
= RHS

QUESTION 6



i) midpoint $(1, 2)$ ①

ii) $m_{AB} = \frac{3-1}{2-0} = \frac{2}{2} = 1$ ①

iii) perp bisector ① for reciprocal gradient
 $y - 2 = -1(x - 1)$ ① for point gradient form
 $y - 2 = -x + 1$
 $x + y - 3 = 0$ ①

iv) $\begin{cases} x + y - 3 = 0 & \textcircled{1} \\ 2x - y - 9 = 0 & \textcircled{2} \end{cases}$ solve simultaneously

① + ② gives

$3x - 12 = 0$

$x = 4$ ① sub into ①

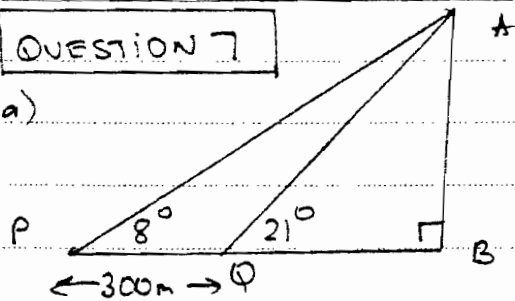
$4 + y - 3 = 0$

$y = -1$ ①

$P(4, -1)$

QUESTION 7

a)



$$i) \angle PAQ = 13^\circ \quad (1)$$

$$ii) \frac{300}{\sin 13^\circ} = \frac{AQ}{\sin 8^\circ} \quad (1)$$

$$AQ = \frac{300 \sin 8^\circ}{\sin 13^\circ}$$

$$AQ = 185.6 \quad (1)$$

$$= 186 \text{ m (nearest m)}$$

$$iii) \sin 21^\circ = \frac{AB}{186} \quad (1)$$

$$AB = 186 \sin 21^\circ$$

$$AB = 67 \text{ m} \quad (1)$$

b) $\tan \theta = -\frac{1}{\sqrt{3}}$ $\begin{array}{c|c} \sqrt{3} & A \\ \hline 1 & C \end{array}$

acute $\theta = 30^\circ$ (1)

$$\therefore \theta = 150^\circ, 330^\circ \quad (1)$$

QUESTION 8

a) $(3, 4) \quad (1)$ lies on $3x - 4y + 7 = 0$

$\therefore$ perp dist $(3, 4)$ to

$$3x - 4y - 3 = 0 \quad (1)$$

$$P = \frac{|9 - 16 - 3|}{\sqrt{9 + 16}} \quad (1)$$

$$= \frac{|-10|}{5}$$

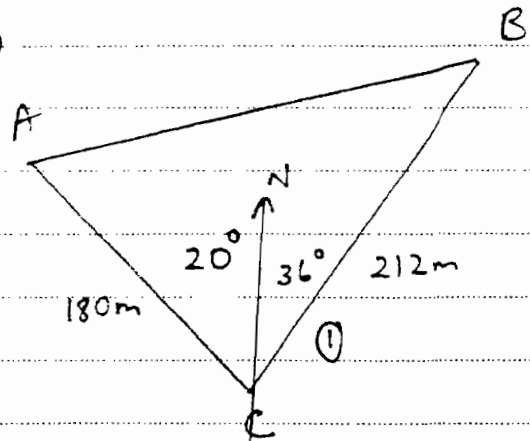
$$P = 2 \text{ units} \quad (1)$$

b) $\operatorname{cosec} 300^\circ = \operatorname{cosec} (360 - 60)$

$$= -\operatorname{cosec} 60^\circ \quad (1)$$

$$= -\frac{2}{\sqrt{3}} \quad (1)$$

c)



$$AB^2 = 180^2 + 212^2 - 2 \times 180 \times 212 \times \cos 56^\circ \quad (1)$$

$$AB = 186 \text{ m} \quad (1)$$